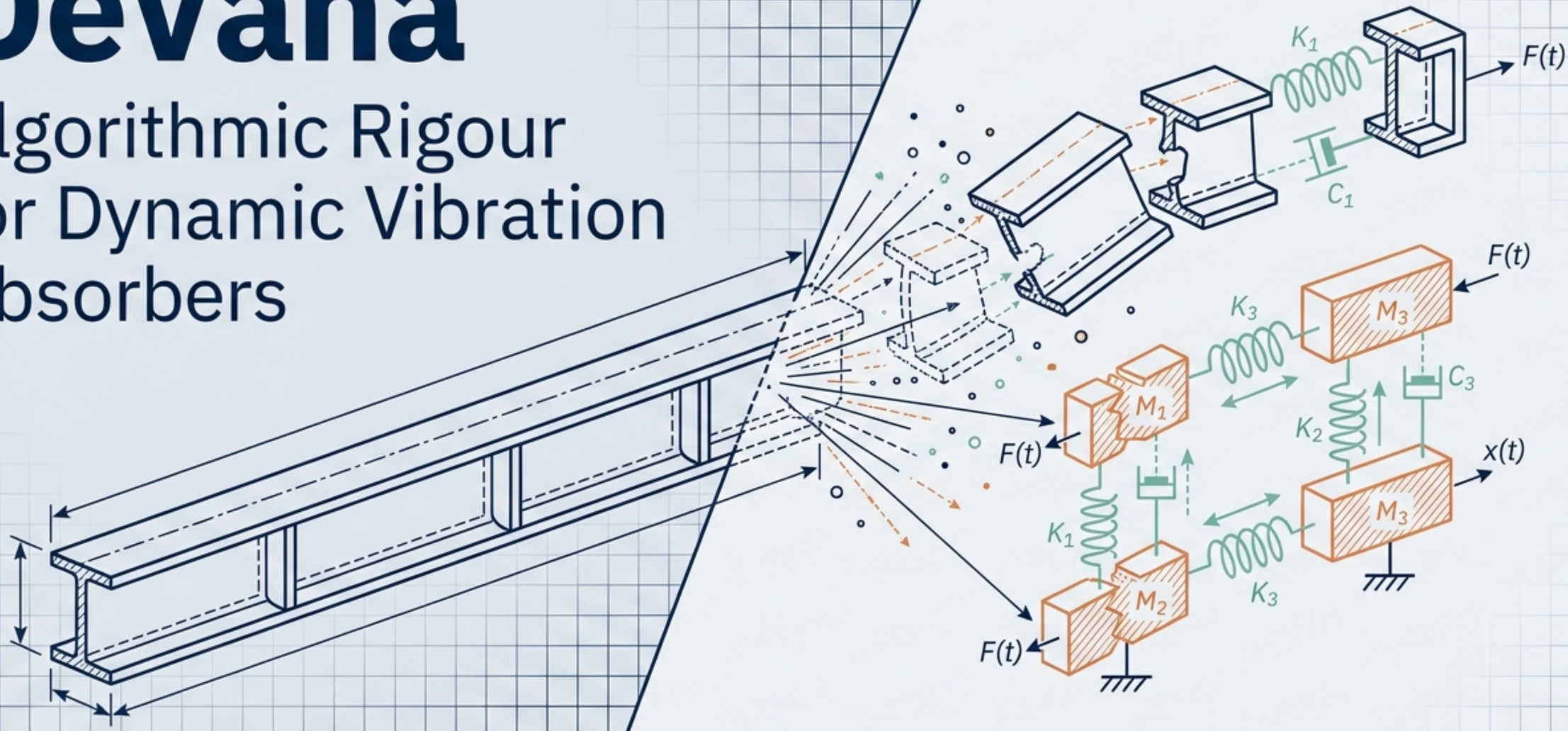
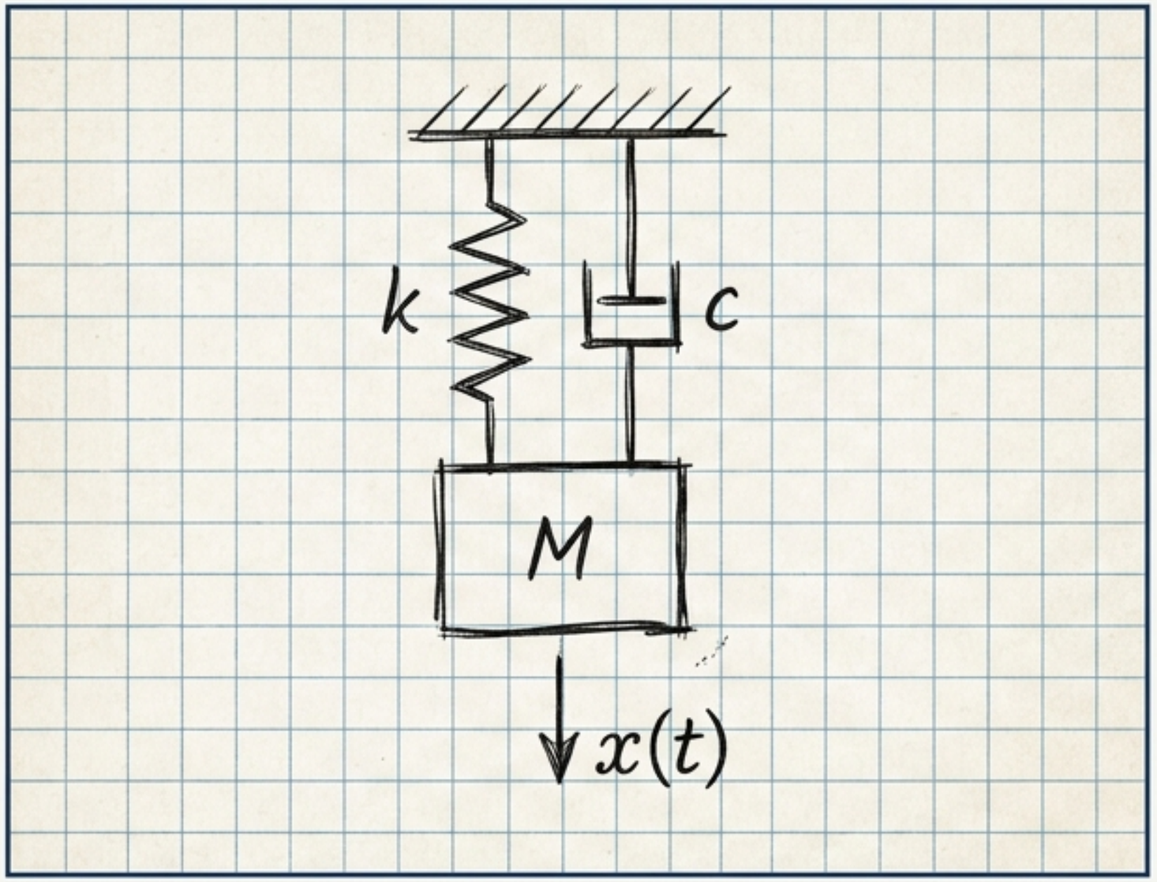
DeVana

Algorithmic Rigour for Dynamic Vibration Absorbers



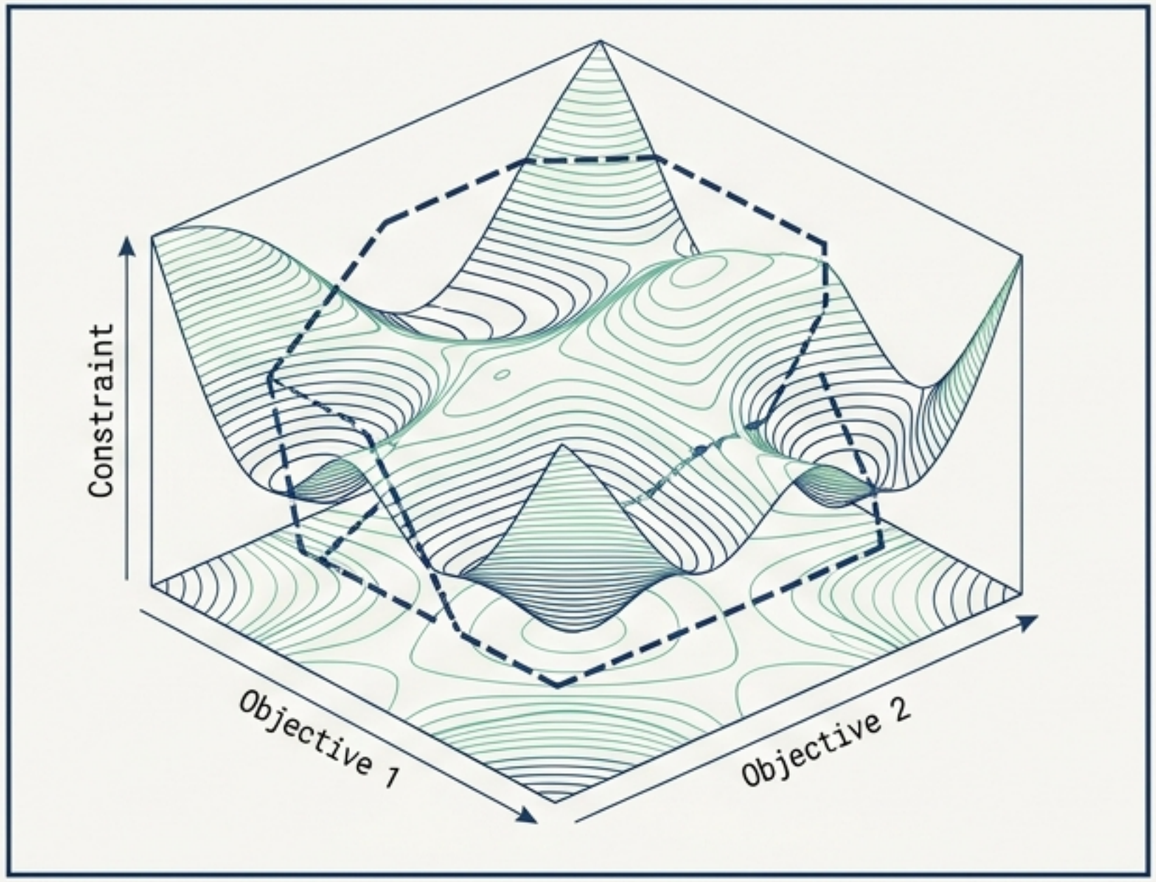
Version: 1.0.0 | Environment: Production | Architecture: Two-Tiered Mixin

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Engineering Intuition

Relies on heuristic knowledge bases and manual trial-and-error. Fails when physical restrictions and multi-criteria objectives multiply the design space.



High-Dimensional Complexity

Modern dynamic vibration absorbers (DVAs) involve competing objectives: cost, weight, component complexity, and vibration dampening across vast, fully-coupled mathematical spaces.

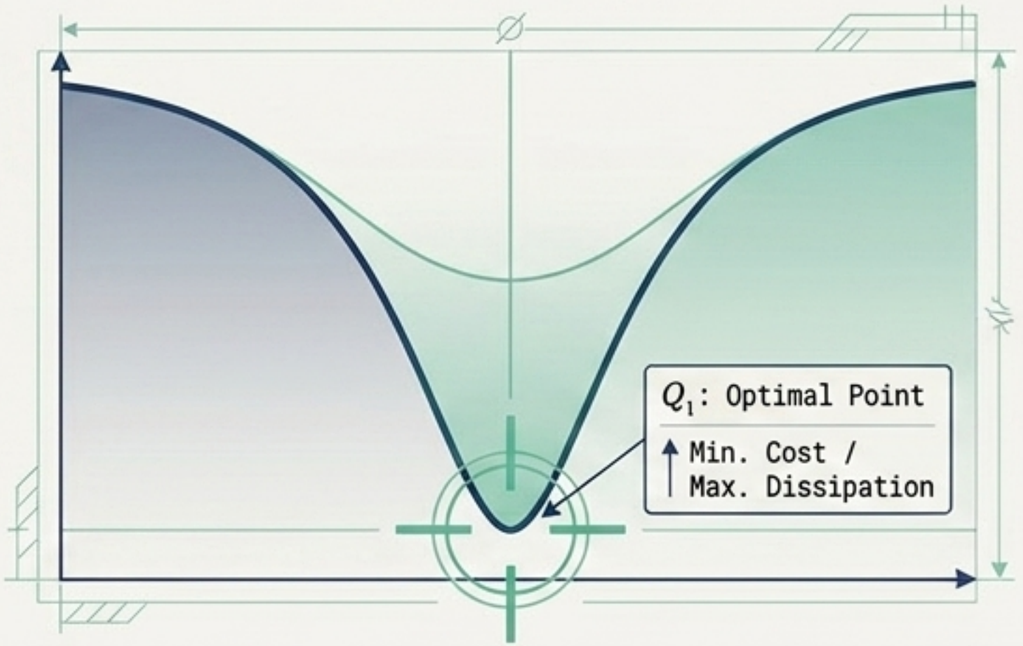
DeVana shifts DVA design from experience-based art to rigorous algorithmic science.

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DeVana Engine

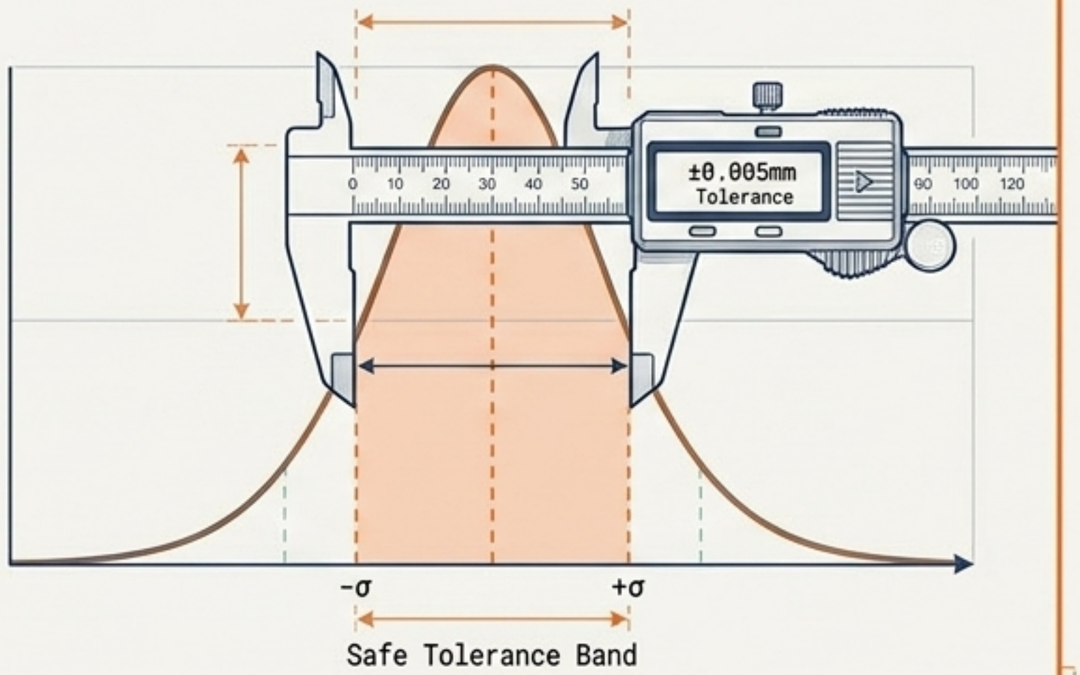
Pathway 1: The Optimal Topology (Q_1)

What is the absolute best combination of components? Minimising active parameters, reducing cost, and maximising energy dissipation to find the singular optimal configuration.



Pathway 2: The Safe Ranges (Q_2)

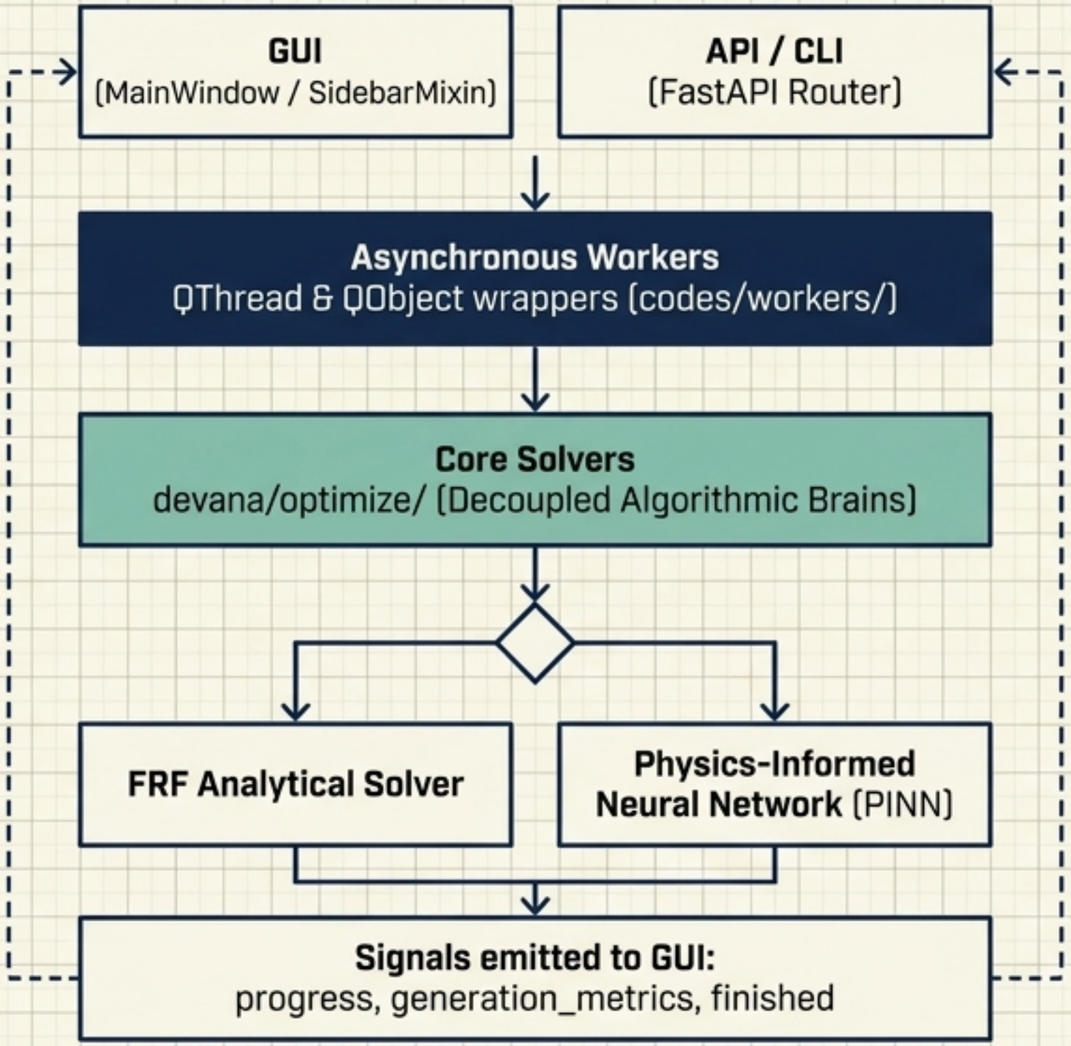
Given real-world manufacturing cannot produce infinite-precision continuous values, what are the statistically safe, reliable bounds for each parameter?



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Software Pipeline Architecture

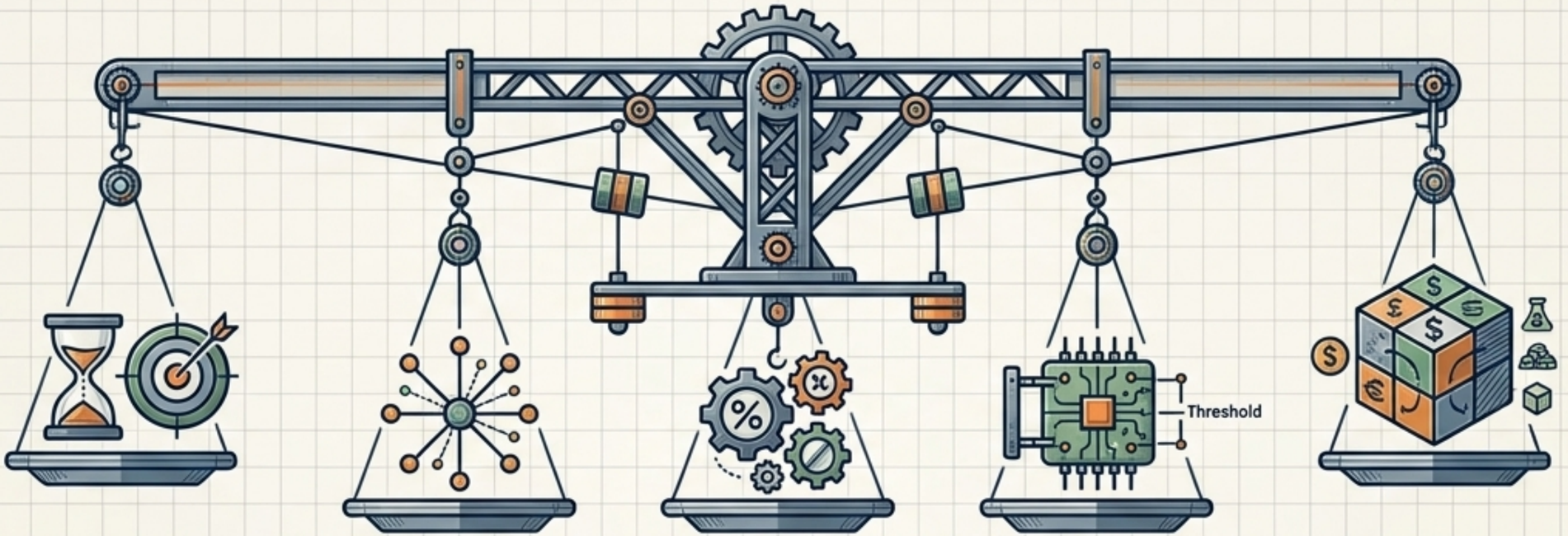
Architectural flowchart mapping top-to-bottom data execution, showing environment splitting, workers, and solvers.



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Modern Engineering Folio

Universal Fitness Formulation: The Five Pillars



Performance ($O_{primary}$): Distance from the normalised ideal singular response.

Sparsity ($P_{sparsity}$): L_1 regularisation penalising complexity to leave a sparse topology.

Accuracy (P_{error}): Scaled sum of percentage deviations across target masses.

Activation ($P_{activation}$): Hard penalty threshold clamping active variables.

Cost (C_{cost}): Multi-objective cost-benefit ratio (material and manufacturing).

Insight: For multi-objective runs (NSGA-II, MOGA), these pillars are decoupled into orthogonal objectives tracking Pareto fronts and Hypervolume metrics.

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Diagnostic Matrix: Algorithmic Brains & Mechanisms

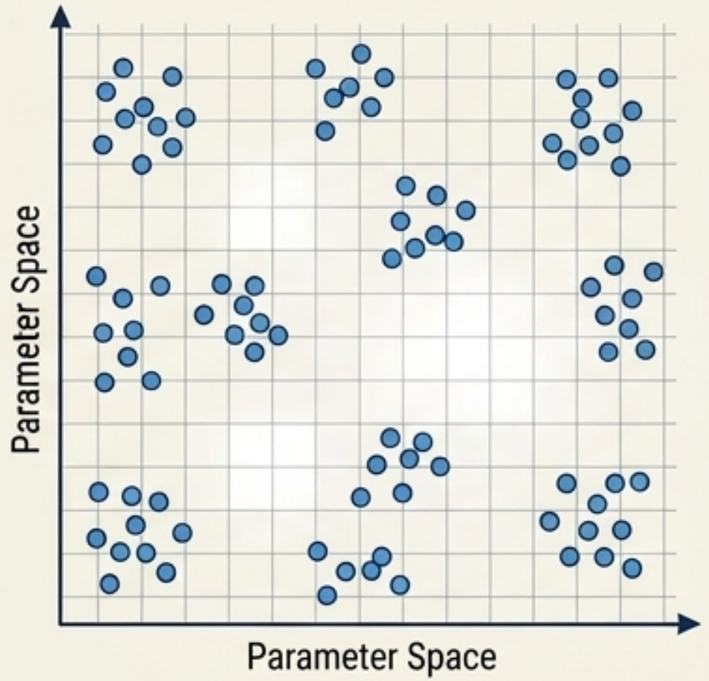
Detailed comparison of core algorithms, use cases, and adaptive control strategies.

Algorithm	Type	Use Case	Key Features	Adaptive Mechanisms
GA	Evolutionary	Single-Objective Core	Multiple seeding, Smart gradient mutation	ML Bandit (UCB), RL Q-Learning, Heuristic rates
NSGA-II / MOGA	Evolutionary	Multi-Objective	Hypervolume tracking, Pareto fronts, f_1, f_2, f_3 decoupling	Simulated Binary Crossover (SBX), Polynomial Mutation
PSO	Swarm	Continuous Search	Velocity clamping, Topological neighbourhoods	Adaptive inertia (w) and acceleration (c_1, c_2)
CMA-ES	Probabilistic	Non-linear/Non-convex	Updates multivariate normal distribution	ML/RL σ step-size controllers
DE	Evolutionary	Robust Global Search	Strategies: rand/1, best/1, current-to-best/1	Adaptive F and CR (JADE, SaDE)
RL	Policy Gradient	Direct Policy Generation	DDPG approach, Experience replay buffer	Epsilon-greedy exploration decay
AdaVEA	Specialised	Bounded Sparsity	Biases toward lower bounds (L_0 sparsity)	Time-decaying crossover/mutation rates

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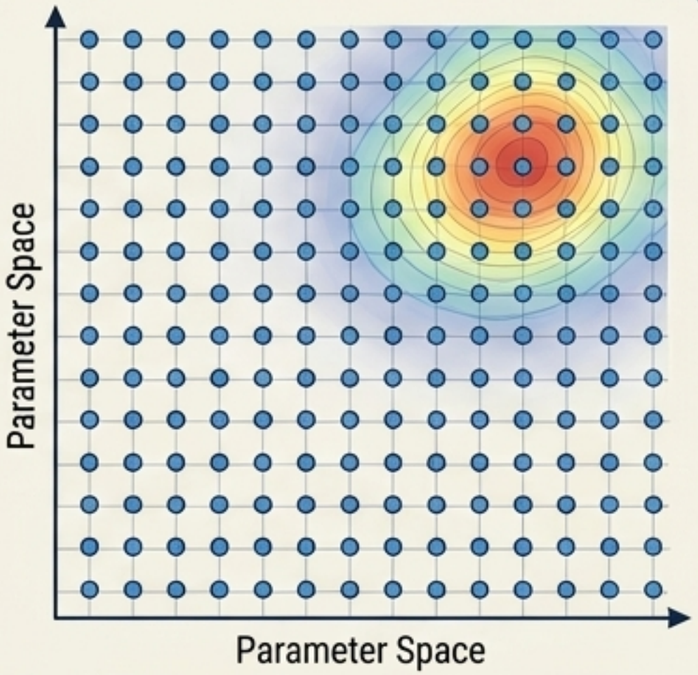
DeVana Arsenal: Advanced Seeding Strategies

Standard Approach: Random Uniform



Standard pseudo-random generation risks local minima trapping and wasted computational cycles on unpromising space.

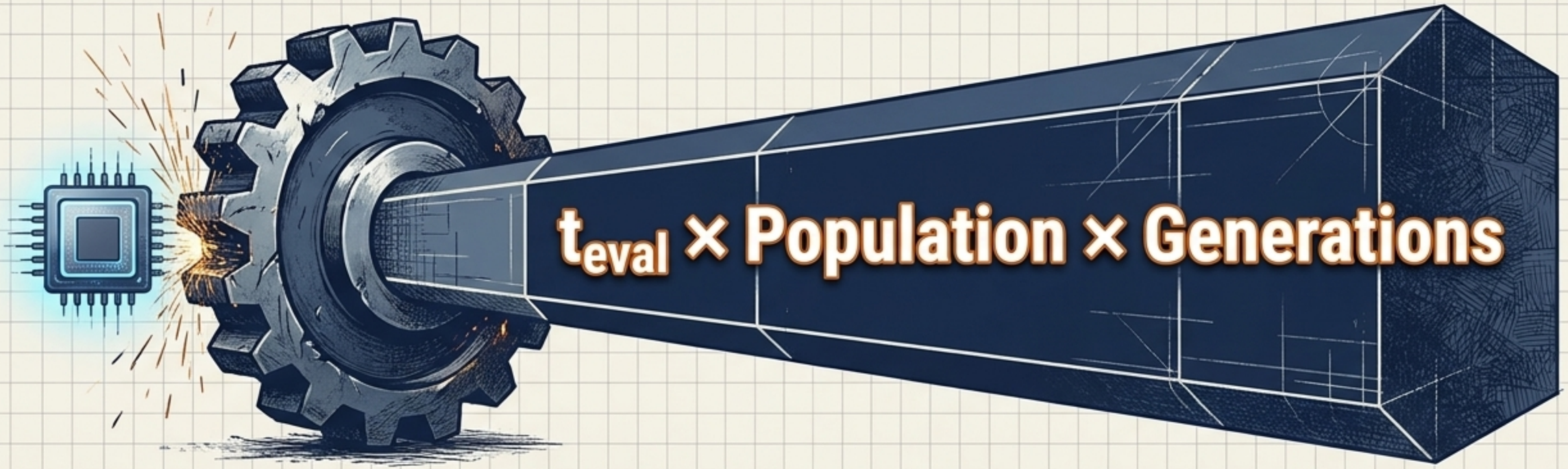
DeVana Arsenal: Intelligent Seeding



- **QMC/Sobol & LHS**: Low-discrepancy sequences ensuring uniform mathematical stratification.
- **Neural UCB/EI**: Online MLP ensembles predict promising regions using Upper Confidence Bound and Expected Improvement.
- **Memory Replay**: Loads historical JSON caches of excellent solutions, injecting them with intentional jitter.

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Computational Friction: The Scaling Burden of Exactness

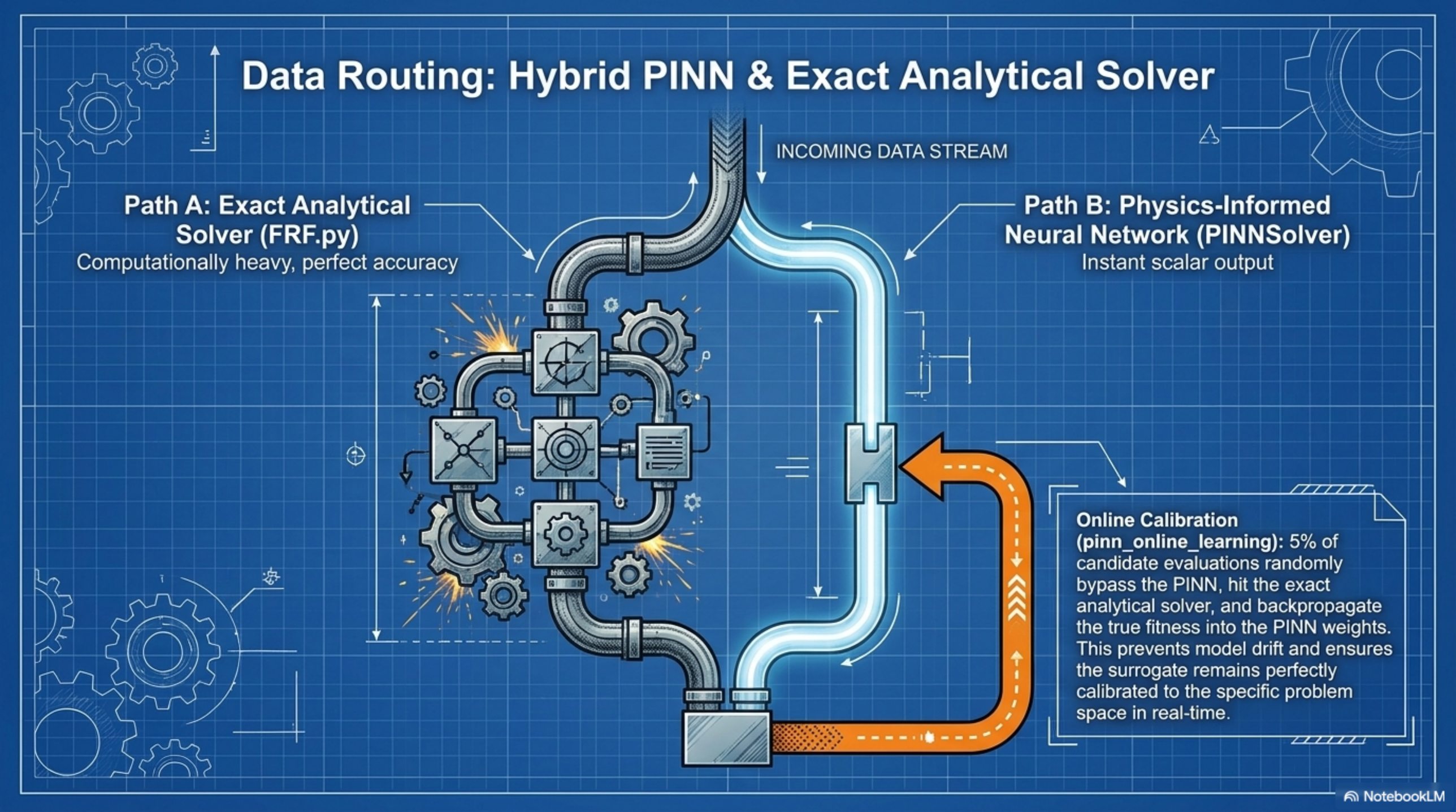


The Friction: To evaluate thousands of parameters across hundreds of generations, the core Frequency Response Function solver must perform complex matrix inversion and zero-DOF elimination repeatedly.

Analytical exactness is computationally heavy. Optimisation requires a paradigm shift in execution speed to remain viable for rapid engineering iteration.

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Data Routing: Hybrid PINN & Exact Analytical Solver



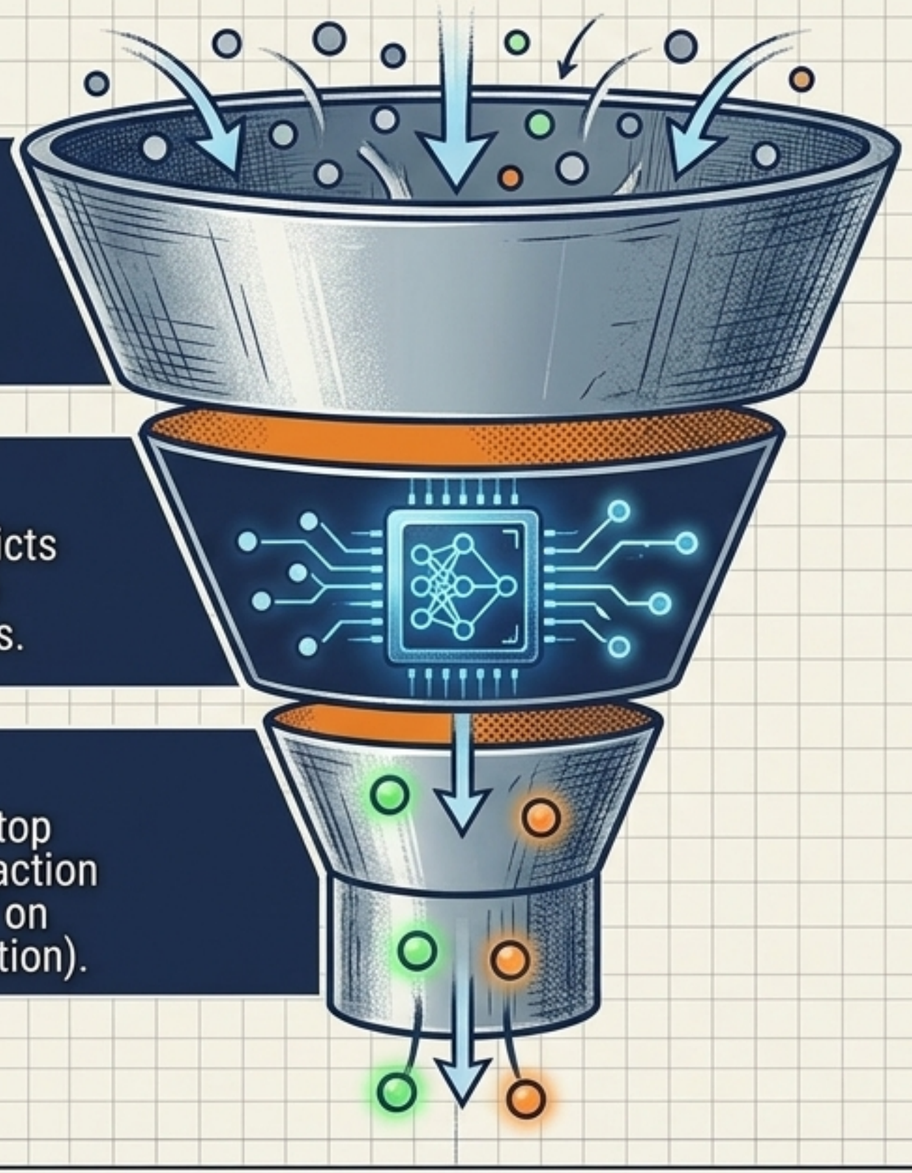
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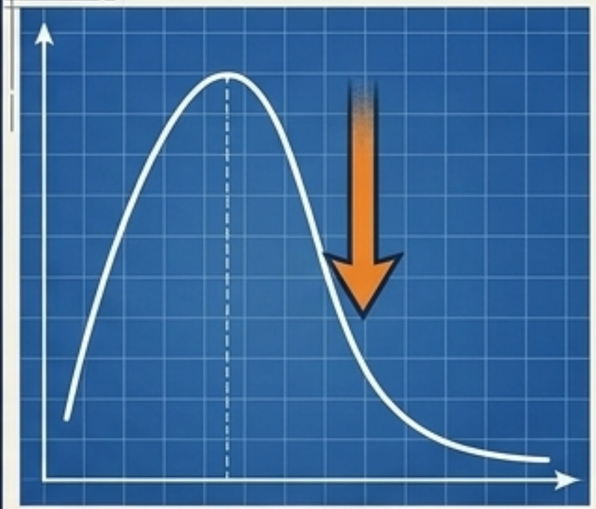
Surrogate-Assisted Optimization: Accelerating the Search.

Over-Sampling
The Genetic Algorithm generates an invalid offspring pool 2x to 5x larger than required.

Neural Prediction
The Neural Surrogate instantly predicts fitness for the entire expanded pool without running physics calculations.

Smart Selection
The algorithm explicitly selects the top N individuals (Exploitation) and a fraction of the most novel individuals based on distance to the training set (Exploration).





Smart Mutation: Takes a gradient-guided descent step derived directly from the surrogate's local gradient, rather than pure Gaussian guesswork.

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Reinforcement Learning Cycle: Agent-Environment Loop

The Agent
(Policy Representation)



Maps outputs securely into continuous parameter bounds via a Sigmoid activation function, avoiding the discretisation errors of tabular Q-learning.

Sobol Pre-ranking

Sobol Sensitivity Analysis runs before training to reorder the policy array, forcing the agent to prioritise the most impactful physical variables first.

Exploration Noise

Injection of Gaussian noise $\mathcal{N}(0, \sigma)$, parametrised by an exponentially decaying ϵ.

Experience Replay Buffer

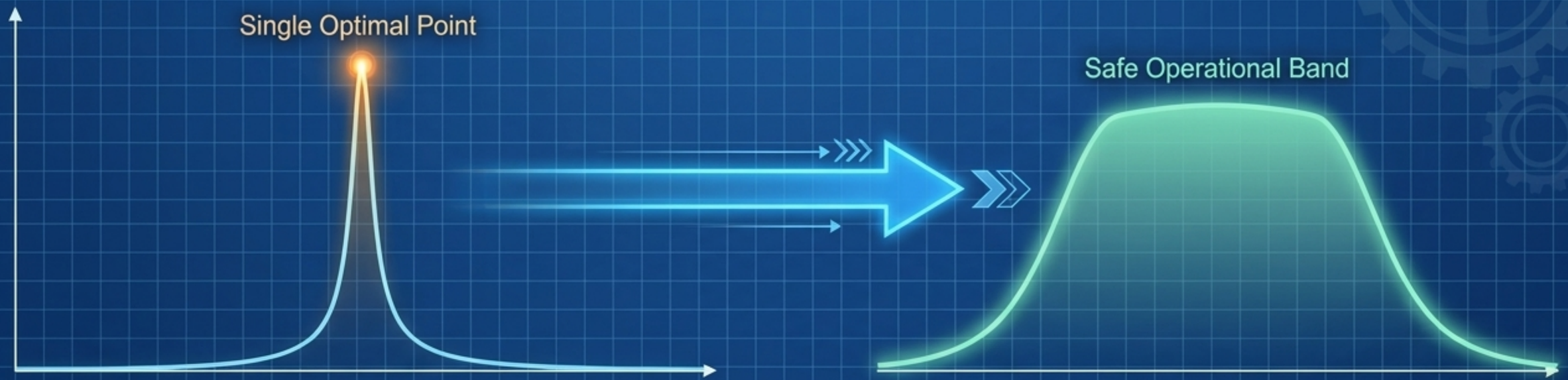


Stores (a_t, F_t) batches. Samples batches to compute the Advantage ($A = -F_t$) to trigger the Policy Gradient Update ($\nabla\theta$).

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DeVana: From Point Optimisation to Range Definition



Step 1: Strict Statistical Extraction

DeVana filters the results to the Top 10% and Median 5% of the Pareto optimal solutions.

Step 2: Boundary Definition

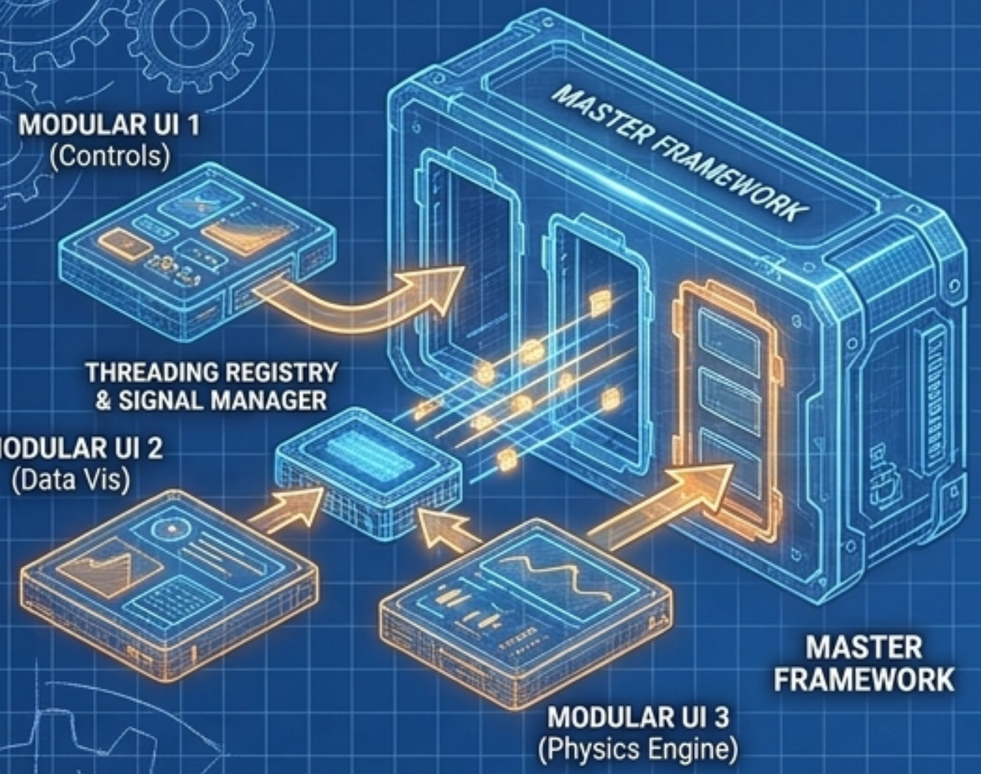
Identifies the continuous tolerance limits where any combination chosen within those bounds guarantees the required vibration dampening.

Engineering Value

Redefining optimisation: Ranges over Points.
Bridges the gap between theoretical multi-DOF equations of motion and actual factory-floor assembly tolerances.

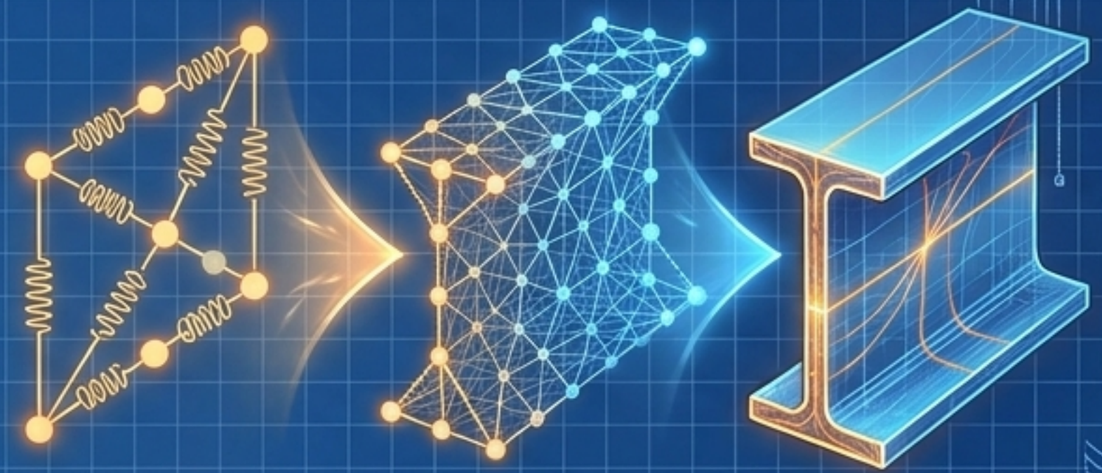
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The Mixin Revolution



The codebase mirrors the engineering: modular, decoupled, and extensible. Threading registries manage signals between UI and physics workers seamlessly.

The Continuous Horizon



DISCRETE DIAGRAM
(Dots & Springs)

DISCRETE-TO-CONTINUOUS TRANSITION

CONTINUOUS 3D STRUCTURAL BEAM
(FEM)

The Next Frontier: Feeding the extracted Safe Ranges directly back into integrated FEM continuous modules for an end-to-end pipeline—from continuous structures to discrete optimisation, and back to continuous validation.

DeVana: Built with rigorous math for the Structural Dynamics Community.

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